

Evaluate limits using Maclaurin's series

Basic skills

Q1. a) $x \sin x$ ✓ $\sin x$'s series

$$= x \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right)$$

$$= x^2 - \frac{x^4}{6} + \frac{x^6}{120} - \dots$$

b) $\cos^2 x = \left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots \right)^2$

$$= 1 - \frac{x^2}{2} - \frac{x^2}{2} + \frac{x^4}{24} + \frac{x^4}{24} + \left(-\frac{x^2}{2} \right)^2 + \dots$$

$$= 1 - x^2 + \frac{x^4}{3} + \dots$$

c) $e^{5x} \sin x = \left(1 + 5x + \frac{25x^2}{2} + \dots \right) \left(x - \frac{x^3}{6} + \dots \right)$

$$= x + 5x^2 + \frac{25}{2}x^3 - \frac{x^3}{6}$$

$$= x + 5x^2 - \frac{37}{3}x^3 + \dots$$

$$\begin{aligned}
 d) \quad (3-x)^{-1} &= 3^{-1} \left(1 - \frac{x}{3}\right)^{-1} \\
 &= \frac{1}{3} \left(1 - 1x - \frac{x}{3} + \frac{-1x-2}{2} \left(-\frac{x}{3}\right)^2 + \dots\right) \\
 &= \frac{1}{3} + \frac{x}{9} + \frac{x^2}{27} + \dots
 \end{aligned}$$

$$\begin{aligned}
 d) \quad \ln^2(1+x) &= \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots\right)^2 \\
 &= x^2 - \frac{x^3}{2} - \frac{x^3}{2} + \frac{x^4}{4} + \frac{x^4}{3} + \frac{x^4}{3} \\
 &= x^2 - x^3 + \frac{11x^4}{12} + \dots
 \end{aligned}$$

$$\begin{aligned}
 Q2. \quad \frac{5x^3+1}{x+1} &= \frac{5x^2 + \frac{1}{x}}{1 + \frac{1}{x}} \rightarrow \frac{\infty}{1} = \infty \\
 &\text{as } x \rightarrow \infty
 \end{aligned}$$

$$\begin{aligned}
 Q3. \quad a) \quad \lim_{x \rightarrow \infty} \frac{x^2 + 5x}{2x^2 - 6x - 2} \\
 = \lim_{x \rightarrow \infty} \frac{1 + \frac{5}{x}}{2 - \frac{6}{x} - \frac{2}{x^2}} = \frac{1}{2}
 \end{aligned}$$

$$b) \lim_{x \rightarrow 0} \frac{x^2 + 5x}{2x^2 - 6x - 2} = \frac{0}{-2} = 0$$

$$c) \lim_{x \rightarrow \infty} \frac{2x^3 - 2}{x^6 - 2} = \lim_{x \rightarrow \infty} \frac{\frac{2}{x^3} - \frac{2}{x^6}}{1 - \frac{2}{x^6}} = \frac{0}{1} = 0$$

$$d) \lim_{x \rightarrow 0} \frac{2x^3 - 2}{x^6 - 2} = \frac{-2}{-2} = 1$$

$$e) \lim_{x \rightarrow 0} \frac{5x + \sqrt{x}}{x + 1} = \frac{0}{1} = 0$$

$$f) \lim_{x \rightarrow \infty} \frac{(x^5 + 1)^2}{x^6 + 5x - 1} = \lim_{x \rightarrow \infty} \frac{x^{10} (1 + x^{-5})^2}{x^6 + 5x - 1}$$

$$= \lim_{x \rightarrow \infty} \frac{x^4 (1 + x^{-5})^2}{1 + 5x^{-5} - x^{-6}} = \infty$$

Competency skills

$$Q4. f(x) = \frac{(x-4)^2}{x^2 + 4} \rightarrow \frac{0^2}{4^2 + 4} = 0 \text{ as } x \rightarrow \infty$$

$$Q5. \lim_{x \rightarrow 0} \frac{\cos x - 1}{5x^2} = \lim_{x \rightarrow 0} \frac{\overset{\text{series}}{(1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots)} - 1}{5x^2}$$

$$= \lim_{x \rightarrow 0} \frac{-\frac{x^2}{2} + \frac{x^4}{24} - \dots}{5x^2} = \lim_{x \rightarrow 0} \frac{-\frac{1}{2} + \frac{x^2}{24} - \dots}{5}$$

$$= \frac{-\frac{1}{2}}{5} = -\frac{1}{10}$$

Q6. $\lim_{x \rightarrow 0} \frac{e^{4x} - (1+4x)}{5x^2}$

$$= \lim_{x \rightarrow 0} \frac{1 + 4x + \frac{16x^2}{2} + \frac{64x^3}{6} + \dots - 1 - 4x}{5x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{16}{2} + \frac{64x}{6} + \dots}{5} = \frac{\frac{16}{2}}{5} = \frac{16}{10}$$

Q7. a) $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{x - \frac{x^3}{6} + \dots}{x}$

$$= \lim_{x \rightarrow 0} \left(1 - \frac{x^2}{6} + \dots\right) = 1$$

b) $\lim_{x \rightarrow 0} \frac{x^2 + 2}{e^x} = \frac{2}{1}$ ← no need for series

$$c) \lim_{x \rightarrow \infty} \frac{x^2+2}{e^x} = \lim_{x \rightarrow \infty} \frac{x^2+2}{1+x+\frac{x^2}{2}+\frac{x^3}{6}+\frac{x^4}{24}+\dots}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{1}{x} + \frac{2}{x^3}}{\frac{1}{x^3} + \frac{1}{x^2} + \frac{1}{2x} + \frac{1}{6} + \frac{x}{24} + \dots}$$

no problem $\rightarrow \frac{0}{\infty} = 0$

$$d) \lim_{x \rightarrow 0} \frac{\sqrt{2-x}+4}{x} = \lim_{x \rightarrow 0} \frac{\sqrt{2(1-\frac{x}{2})}+4}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{2(1-\frac{1}{4}x + \frac{\frac{1}{2}x - \frac{1}{2}}{2}(\frac{-x}{2})^2 + \dots)} + 4}{x}$$

$$= \frac{\sqrt{2}}{0} = \infty$$

$$e) \lim_{x \rightarrow \infty} \frac{\ln(1+x)}{2x+x^2} = \lim_{x \rightarrow \infty} \frac{x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots}{2x+x^2}$$

$$= \frac{\frac{1}{x} - \frac{1}{2} + \frac{x}{3}}{\frac{2}{x} + 1} = \frac{\infty}{1} = \infty$$

$$f) \lim_{x \rightarrow 0} \frac{(x+5)\sin x}{\cos x} = \frac{0}{1} = 0$$

no problem

$$g) \lim_{x \rightarrow 0} \frac{2 - e^x \sin x}{\cos(x + \frac{\pi}{2})}$$

↓ may consider this

$$= \lim_{x \rightarrow 0} \frac{2 - (1 + x + \frac{x^2}{2} + \dots)(x - \frac{x^3}{6} + \dots)}{1 - \frac{(x + \frac{\pi}{2})^2}{2} + \dots}$$

messy

issue is

$$\frac{(\frac{\pi}{2})^2}{2} + \frac{(\frac{\pi}{2})^3}{6} + \dots$$

infinite sum

Instead

$$\lim_{x \rightarrow 0} \frac{2 - e^x \sin x}{- \sin x}$$

$$= \lim_{x \rightarrow 0} \frac{2 - (x + x^2 - \dots)}{(x - \frac{x^3}{3} + \dots)}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{2}{x} - 1 + x - \dots}{1 - \frac{x^2}{3} + \dots} = \frac{\infty}{1} = \infty$$

Q8. when the function is continuous and defined at limit so not 0 and not ∞

Q9. $\lim_{x \rightarrow 0} \frac{e^x + e^{-x}}{e^x - e^{-x}}$

$$= \lim_{x \rightarrow 0} \frac{(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots) + (1 - x + \frac{x^2}{2} - \dots)}{(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots) - (1 - x + \frac{x^2}{2} - \dots)}$$

$$= \lim_{x \rightarrow 0} \frac{2 + x^2 + \frac{x^4}{12} + \dots}{2x + \frac{x^3}{3} + \frac{x^5}{60} + \dots}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\frac{2}{x} + x + \dots}{2 + \frac{x^2}{3} + \dots}}{\frac{2}{x} + x + \dots} = \frac{\infty}{2} = \infty$$

As an alternative $f(x) = \frac{e^{2x} + 1}{e^{2x} - 1}$

$$\text{Q10. } \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x) - 2x}{e^{5x} - 1 - 5x}$$

$$= \lim_{x \rightarrow 0} \frac{\ln(1 + (x - \frac{x^3}{6} + \dots)) - x}{e^{5x} - 1 - 5x}$$

$$= \lim_{x \rightarrow 0} \frac{(x - \frac{x^3}{6} + \dots) - \frac{(x - \frac{x^3}{6} + \dots)^2}{2} + \frac{(x - \frac{x^3}{6} + \dots)^3}{3} - 2x}{e^{5x} - 1 - 5x}$$

$$(1 + 5x + \frac{25}{2}x^2 + \frac{125}{6}x^3 + \dots) - 1 - 5x$$

$$= \lim_{x \rightarrow 0} \frac{-\frac{x^3}{6} - \frac{x^2}{2} + \frac{x^3}{3} + \dots}{\frac{25}{2}x^2 + \frac{125}{6}x^3 + \dots}$$

$$= \lim_{x \rightarrow 0} \frac{-\frac{1}{2} + \frac{x}{6} + \dots}{\frac{25}{2} + \frac{125}{6}x + \dots} = \frac{-\frac{1}{2}}{\frac{25}{2}} = -\frac{1}{25}$$

$$Q11. a) f(x) = \ln(\cos x + \sin x)$$

$$= \ln\left(1 - \frac{x^2}{2} + \dots + x - \frac{x^3}{6} + \dots\right)$$

$$= \ln\left(1 + \left(x - \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} - \dots\right)\right)$$

$$= \ln(1 + g(x))$$

$$= g(x) - \frac{(g(x))^2}{2} + \frac{(g(x))^3}{3} - \dots$$

$$= x - \frac{x^2}{2} - \frac{x^3}{6} - \frac{x^2 - \frac{x^3}{2} - \frac{x^3}{2}}{2} + \frac{x^3}{3} + \dots$$

$$= x - x^2 + \frac{2}{3}x^3 - \dots$$

$$b) \lim_{x \rightarrow \infty} \frac{f(x)}{5x^2 - 9} = \lim_{x \rightarrow \infty} \frac{x - x^2 + \frac{2}{3}x^3 - \dots}{5x^2 - 9}$$

$$= \frac{\frac{1}{x} - 1 + \frac{2}{3}x - \dots}{5 - \frac{9}{x^2}}$$

+ might guess is " $\frac{\infty}{5}$ " = ∞ .

However alternating sign causes an issue so not possible (yet)